

ELITE IGCSE ACADEMY

Student Past Paper Solutions

Nov 2024 P2H Worked Solutions

Nov 2024 | P2H

31 questions ■ question image followed by worked solution

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STUDENT EDITION

Worked answers included

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PAST PAPER QUESTION**Q#: 1****Marks: 3**TOPIC: **Algebraic Proof**

Nov 2024 P2H - Nov 2024 | P2H

1 Show that $1\frac{5}{7} \times 2\frac{3}{16} = 3\frac{3}{4}$

(Total for Question 1 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 1****Marks: 3****TOPIC: Algebraic Proof**

Nov 2024 P2H - Nov 2024 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Evaluate fraction**

$$1\frac{5}{7} = \frac{12}{7}, \quad 2\frac{3}{16} = \frac{35}{16}$$

$$1\frac{5}{7} \times 2\frac{3}{16} = \frac{12}{7} \times \frac{35}{16}$$

$$= \frac{420}{112} = \frac{15}{4} = 3\frac{3}{4}$$

FINAL ANSWER

$$3\frac{3}{4}.$$

EA

PAST PAPER QUESTION**Q#: 2****Marks: 2****TOPIC: Rounding, Estimation & Bounds**

Nov 2024 P2H - Nov 2024 | P2H

2 The length of a table is measured as 1.4 metres correct to one decimal place.

(a) Write down the upper bound of the length of the table.

..... metres

(1)

(b) Write down the lower bound of the length of the table.

..... metres

(1)

(Total for Question 2 is 2 marks)

EA

PAST PAPER SOLUTION**Q#: 2****Marks: 2****TOPIC: Rounding, Estimation & Bounds**

Nov 2024 P2H - Nov 2024 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. The length is m correct**

The length is 1.4 m correct to one decimal place.

2. Half of is

Half of 0.1 is 0.05.

$$1.4 + 0.05 = 1.45$$

$$1.4 - 0.05 = 1.35$$

FINAL ANSWER

(a) 1.45 m, (b) 1.35 m

PAST PAPER QUESTION**Q#: 3****Marks: 3****TOPIC: Right-Angled Triangles - Pythagoras & Trigonometry**

Nov 2024 P2H - Nov 2024 | P2H

- 3 The diagram shows triangle PQR

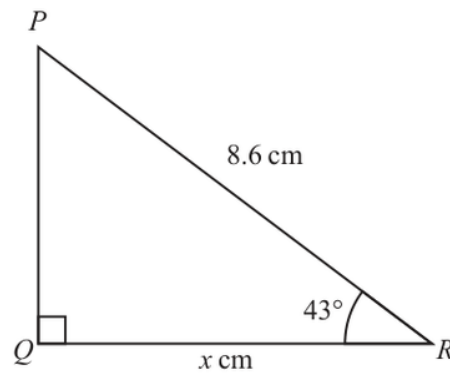


Diagram **NOT**
accurately drawn

Work out the value of x
Give your answer correct to one decimal place.

$x =$

(Total for Question 3 is 3 marks)

PAST PAPER SOLUTION**Q#: 3****Marks: 3****TOPIC:** Right-Angled Triangles - Pythagoras & Trigonometry

Nov 2024 P2H - Nov 2024 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use trigonometry**

In the right-angled triangle, x is adjacent to the 43° angle.

$$\cos 43^\circ = \frac{x}{8.6}$$

$$x = 8.6 \cos 43^\circ = 6.289 \dots$$

FINAL ANSWER

6.3 cm.

EA

PAST PAPER QUESTION**Q#: 4****Marks: 5****TOPIC: Percentages**

Nov 2024 P2H - Nov 2024 | P2H

- 4 N is a number.
17% of N is 357

(a) Work out the value of N

$$N = \dots\dots\dots (2)$$

In 2019, the population of a village was 650
In 2020, the population of the village was 806

(b) Work out the percentage increase in the population.

$$\dots\dots\dots \% (3)$$

(Total for Question 4 is 5 marks)

PAST PAPER SOLUTION**Q#: 4****Marks: 5****TOPIC: Percentages**

Nov 2024 P2H - Nov 2024 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Calculate percentage**

For part (a),

$$17\% \text{ of } N = 357$$

$$0.17N = 357$$

$$N = \frac{357}{0.17} = 2100$$

2. Calculate percentage

For part (b), the increase in population is

$$806 - 650 = 156$$

3. Percentage increase

Percentage increase:

$$\frac{156}{650} \times 100 = 24$$

FINAL ANSWER(a) $N = 2100$, (b) 24%

PAST PAPER QUESTION**Q#: 5****Marks: 4****TOPIC: Probability Toolkit**

Nov 2024 P2H - Nov 2024 | P2H

- 5** Cody has a biased 5-sided spinner, numbered 1, 2, 3, 4, 5

The table gives the probabilities that when the spinner is spun it will land on 2 or on 3 or on 5

Number	1	2	3	4	5
Probability		0.14	0.17		0.21

The probability that the spinner will land on 1 is the same as the probability that the spinner will land on 4

Cody is going to spin the spinner 400 times.

Work out an estimate for the number of times the spinner will land on 4

(Total for Question 5 is 4 marks)

PAST PAPER SOLUTION**Q#: 5****Marks: 4****TOPIC: Probability Toolkit**

Nov 2024 P2H - Nov 2024 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Calculate probability**

The known probabilities add to

$$0.18 + 0.17 + 0.23 = 0.58$$

2. Calculate probability

So the probabilities for 1 and 4 add to

$$1 - 0.58 = 0.42$$

3. Calculate probability

They are equal, so

$$P(4) = 0.42 \div 2 = 0.21$$

4. Expected number of s

Expected number of 4s:

$$400 \times 0.21 = 84$$

FINAL ANSWER

84.

PAST PAPER QUESTION**Q#: 6****Marks: 3****TOPIC: Volume & Surface Area**

Nov 2024 P2H - Nov 2024 | P2H

- 6 The diagram shows a solid triangular prism.

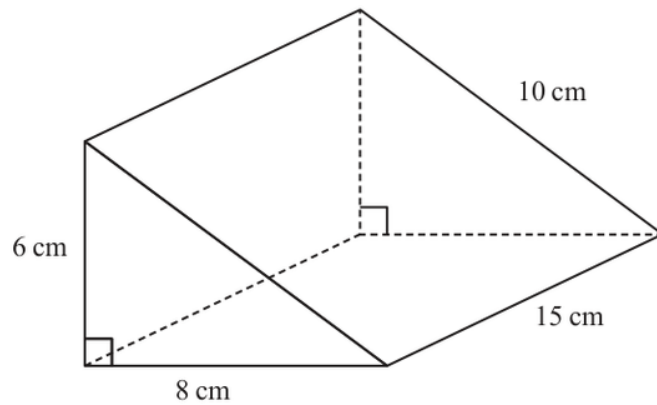


Diagram **NOT**
accurately drawn

Work out the **total** surface area of the triangular prism.

..... cm^2

(Total for Question 6 is 3 marks)

PAST PAPER SOLUTION**Q#: 6****Marks: 3****TOPIC: Volume & Surface Area**

Nov 2024 P2H - Nov 2024 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. The triangular end is a**

The triangular end is a right triangle with sides 6, 8, and 10.

2. Area of the two triangular

Area of the two triangular ends:

$$2 \left(\frac{1}{2} \times 6 \times 8 \right) = 48$$

3. Area of the three rectangles

Area of the three rectangles:

$$15(6 + 8 + 10) = 360$$

4. Total surface area

Total surface area:

$$48 + 360 = 408$$

FINAL ANSWER

408 cm².

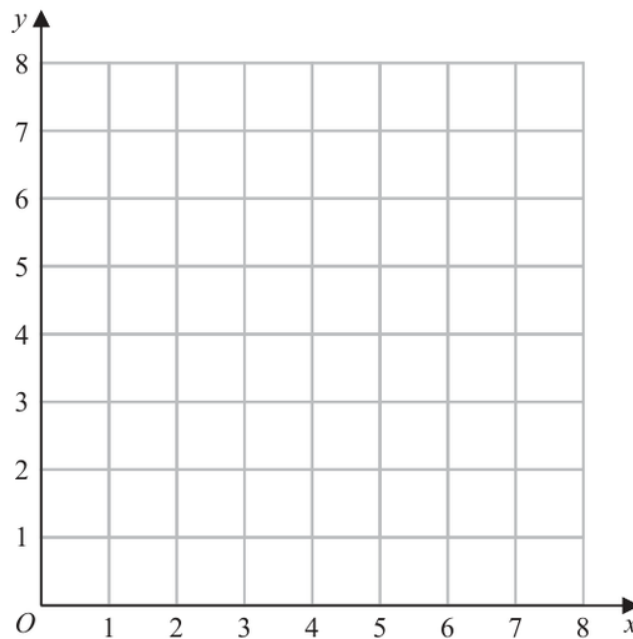
PAST PAPER QUESTION**Q#: 7****Marks: 4****TOPIC: Graphing Inequalities**

Nov 2024 P2H - Nov 2024 | P2H

7 (a) On the grid, draw the straight line with equation

(i) $x = 3$ (ii) $y = 1$ (iii) $x + y = 7$

Label each line with its equation.



(3)

(b) Show, by shading on the grid, the region that satisfies all three of the inequalities

$x \geq 3$ $y \geq 1$ $x + y \leq 7$

Label the region **R**

(1)

(Total for Question 7 is 4 marks)

PAST PAPER SOLUTION

Q#: 7

Marks: 4

TOPIC: **Graphing Inequalities**

Nov 2024 P2H - Nov 2024 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Draw the boundary lines

Draw the boundary lines:

$$x = 3, \quad y = 1, \quad x + y = 7$$

2. The required region is

The required region is

$$x \geq 3$$

$$y \geq 1$$

$$x + y \leq 7$$

FINAL ANSWERShade $x \geq 3$, $y \geq 1$, $x + y \leq 7$.

PAST PAPER QUESTION**Q#: 8****Marks: 3**TOPIC: **Statistics Toolkit**

Nov 2024 P2H - Nov 2024 | P2H

- 8** Kim puts 4 bananas in a bag.

The mean weight of the 4 bananas in the bag is 145 grams.

Andy puts one more banana into the bag.

The mean weight of the 5 bananas in the bag is 142 grams.

Work out the weight of the banana that Andy puts into the bag.

..... grams

(Total for Question 8 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 8****Marks: 3****TOPIC: Statistics Toolkit**

Nov 2024 P2H - Nov 2024 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. The total weight of the**

The total weight of the first 4 bananas is

$$4 \times 145 = 580$$

2. The total weight of the

The total weight of the 5 bananas is

$$5 \times 142 = 710$$

3. Weight of the banana Andy

Weight of the banana Andy added:

$$710 - 580 = 130$$

FINAL ANSWER

130 g.

PAST PAPER QUESTION**Q#: 9****Marks: 3****TOPIC: Compound Interest & Depreciation**

Nov 2024 P2H - Nov 2024 | P2H

- 9 Nisha invests 20 000 euros for 3 years in a savings account.
She gets 3.5% per year compound interest.

Work out how much money Nisha will have in her savings account
at the end of the 3 years.

Give your answer correct to the nearest euro.

..... euros

(Total for Question 9 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 9

Marks: 3

TOPIC: **Compound Interest & Depreciation**

Nov 2024 P2H - Nov 2024 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. The multiplier for compound interest

The multiplier for 3.5% compound interest is

$$1.035$$

2. Use compound interest

After 3 years,

$$20000(1.035)^3 = 22174.3575$$

3. Use compound interest

Correct to the nearest euro,

$$22174.3575 \approx 22174$$

FINAL ANSWER

22174 euros

PAST PAPER QUESTION

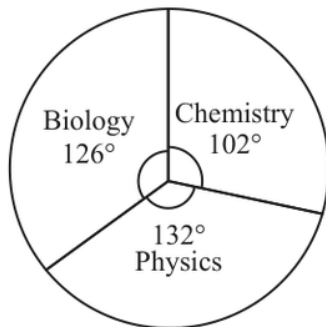
Q#: 10 Marks: 5

TOPIC: **Statistics Toolkit**

Nov 2024 P2H - Nov 2024 | P2H

10 All the students in year 10 and all the students in year 11 named their favourite science subject from Biology, Chemistry and Physics.

The pie chart shows information about the results for the year 10 students.
The table shows information about the results for the year 11 students.



Pie chart for year 10

science subject	number of students
Biology	$3x + 6$
Chemistry	$5x + 8$
Physics	$7x - 9$

Table for year 11

There are 300 students in year 10
There are 320 students in year 11

More students in year 10 than in year 11 said Biology was their favourite science subject.

How many more?

(Total for Question 10 is 5 marks)

PAST PAPER SOLUTION**Q#: 10****Marks: 5****TOPIC: Statistics Toolkit**

Nov 2024 P2H - Nov 2024 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Evaluate fraction**For year 10, Biology is 126° out of 360° .

$$\frac{126}{360} \times 300 = 105$$

2. Calculate value

For year 11:

$$(3x + 6) + (5x + 8) + (7x - 9) = 320$$

$$15x + 5 = 320$$

$$x = 21$$

3. Year 11 Biology

Year 11 Biology:

$$3(21) + 6 = 69$$

4. Difference

Difference:

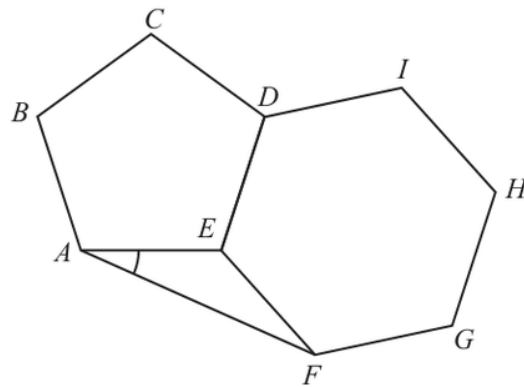
$$105 - 69 = 36$$

FINAL ANSWER

36 more students.

PAST PAPER QUESTION**Q#: 11****Marks: 5****TOPIC: Angles in Polygons & Parallel Lines**

Nov 2024 P2H - Nov 2024 | P2H

11Diagram **NOT**
accurately drawn*ABCDE* is a regular pentagon.*DEFGHI* is a regular hexagon.*AF* is a straight line.Work out the size of angle *EAF*

o

(Total for Question 11 is 5 marks)

PAST PAPER SOLUTION**Q#: 11****Marks: 5****TOPIC: Angles in Polygons & Parallel Lines**

Nov 2024 P2H - Nov 2024 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. A regular pentagon has interior**

A regular pentagon has interior angle

$$108^\circ$$

2. A regular hexagon has interior

A regular hexagon has interior angle

$$120^\circ$$

3. Use trigonometryAt E , the angle inside triangle AEF is

$$360^\circ - 108^\circ - 120^\circ = 132^\circ$$

4. Use trigonometryAlso $AE = ED$ and $ED = EF$, so

$$AE = EF$$

5. Use trigonometryTriangle AEF is isosceles, so the base angles are equal:

$$\angle EAF = \frac{180^\circ - 132^\circ}{2} = 24^\circ$$

FINAL ANSWER 24° .

PAST PAPER QUESTION**Q#: 12****Marks: 7****TOPIC: Rearranging Formulas**

Nov 2024 P2H - Nov 2024 | P2H

12 (a) Solve $\frac{3x+2}{5} - \frac{2x+1}{3} = x$

Show clear algebraic working.

$$x = \dots\dots\dots$$

(3)

(b) Make c the subject of the formula $f = \sqrt{\frac{a+bc}{c-d}}$

$$\dots\dots\dots$$

(4)

(Total for Question 12 is 7 marks)

EA

PAST PAPER SOLUTION**Q#: 12****Marks: 7****TOPIC: Rearranging Formulas**

Nov 2024 P2H - Nov 2024 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. (a)**

(a)

$$\frac{3x+2}{5} - \frac{2x+1}{3} = x$$

2. Multiply by 15

Multiply by 15:

$$3(3x+2) - 5(2x+1) = 15x$$

$$9x+6 - 10x-5 = 15x$$

$$1 = 16x$$

$$x = \frac{1}{16}$$

3. (b)

(b)

$$f = \sqrt{\frac{a+bc}{c-d}}$$

$$f^2 = \frac{a+bc}{c-d}$$

$$f^2(c-d) = a+bc$$

$$c(f^2 - b) = a + df^2$$

FINAL ANSWER

$$(a) x = \frac{1}{16}, (b) c = \frac{a+df^2}{f^2-b}$$

PAST PAPER QUESTION

Q#: 13 Marks: 6

TOPIC: Cumulative Frequency Diagrams

Nov 2024 P2H - Nov 2024 | P2H

13 The frequency table gives information about the weights, in kilograms, of 60 parcels in a delivery van.

Weight (w kilograms)	Frequency
$0 < w \leq 1$	4
$1 < w \leq 2$	15
$2 < w \leq 3$	20
$3 < w \leq 4$	11
$4 < w \leq 5$	6
$5 < w \leq 6$	4

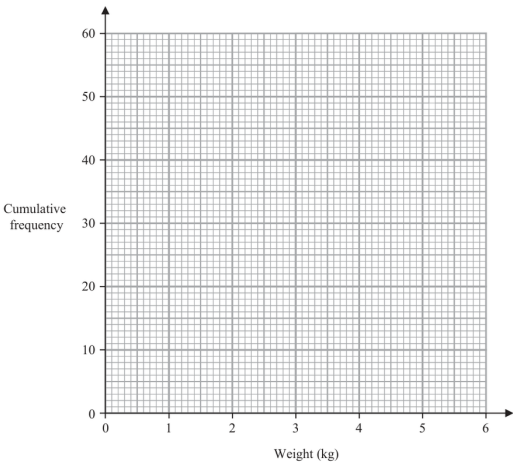
(a) Complete the cumulative frequency table.

Weight (w kilograms)	Cumulative frequency
$0 < w \leq 1$	
$0 < w \leq 2$	
$0 < w \leq 3$	
$0 < w \leq 4$	
$0 < w \leq 5$	
$0 < w \leq 6$	

(1)

(b) On the grid opposite, draw a cumulative frequency graph for your table.

(2)



(c) Use your graph to find an estimate for the median weight of the 60 parcels.

_____ kilograms
(1)

(d) Use your graph to find an estimate for the number of these parcels that weigh more than 3.7 kilograms.

(2)

(Total for Question 13 is 6 marks)

PAST PAPER SOLUTION**Q#: 13****Marks: 6****TOPIC: Cumulative Frequency Diagrams**

Nov 2024 P2H - Nov 2024 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. The cumulative frequencies are**

The cumulative frequencies are

$$4, 19, 39, 50, 56, 60$$

2. The median is the th

The median is the 30th value.

$$2 + \frac{30 - 19}{39 - 19} \times 1 = 2.55$$

3. Use cumulative frequency

At 3.7 kg,

$$F(3.7) \approx 39 + \frac{0.7}{1}(50 - 39) = 46.7$$

4. Use cumulative frequency

So the number heavier than 3.7 kg is

$$60 - 46.7 = 13.3$$

FINAL ANSWER

median about 2.55 kg, and about 13 parcels.

PAST PAPER QUESTION**Q#: 14****Marks: 3****TOPIC: Algebraic Roots & Indices**

Nov 2024 P2H - Nov 2024 | P2H

14 Given that $\frac{3^{2n+3}}{3^4} = 3^3 \times 3^{1-2n}$

find the value of n
Show your working clearly.

$$n = \dots\dots\dots$$

(Total for Question 14 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 14

Marks: 3

TOPIC: **Algebraic Roots & Indices**

Nov 2024 P2H - Nov 2024 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Use index laws

$$\frac{3^{2n+3}}{3^4} = 3^{2n-1}$$

$$3^3 \times 3^{1-2n} = 3^{4-2n}$$

2. Use index laws

So

$$2n - 1 = 4 - 2n$$

$$4n = 5$$

FINAL ANSWER

$$n = \frac{5}{4}$$

PAST PAPER QUESTION**Q#: 15****Marks: 2**TOPIC: **Algebraic Proof**

Nov 2024 P2H - Nov 2024 | P2H

15 Use algebra to show that $0.\dot{7}\dot{6}\dot{3} = \frac{42}{55}$

(Total for Question 15 is 2 marks)

EA

PAST PAPER SOLUTION**Q#: 15****Marks: 2****TOPIC: Algebraic Proof**

Nov 2024 P2H - Nov 2024 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Let**

Let

$$x = 0.7636363\dots$$

2. Calculate value

Then

$$10x = 7.636363\dots$$

3. Calculate value

and

$$1000x = 763.636363\dots$$

4. Subtract

Subtract:

$$990x = 756$$

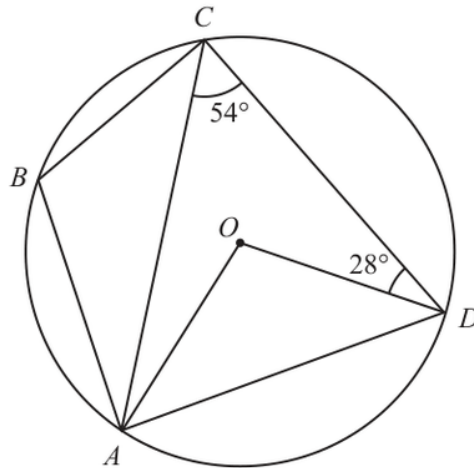
$$x = \frac{756}{990} = \frac{42}{55}$$

FINAL ANSWER

$$0.7636363\dots = \frac{42}{55}$$

PAST PAPER QUESTION**Q#: 16****Marks: 5****TOPIC: Circle Theorems**

Nov 2024 P2H - Nov 2024 | P2H

16 A, B, C and D are points on a circle, centre O Diagram **NOT**
accurately drawn(a) (i) Work out the size of angle AOD
(1)

(ii) Give a reason for your answer to part (a)(i)

.....
(1)(b) Work out the size of angle CAO
(1)(c) Work out the size of angle ABC
(2)**(Total for Question 16 is 5 marks)**

PAST PAPER SOLUTION**Q#: 16****Marks: 5****TOPIC: Circle Theorems**

Nov 2024 P2H - Nov 2024 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. (a) The angle at the**(a) The angle at the centre is twice the angle at the circumference standing on the same arc AD :

$$\angle AOD = 2 \times 54^\circ = 108^\circ$$

2. Use trigonometry(b) Since $OC = OD$, triangle COD is isosceles:

$$\angle OCD = \angle CDO = 28^\circ$$

3. Use trigonometry

So

$$\angle COD = 180^\circ - 28^\circ - 28^\circ = 124^\circ$$

4. Use circle theoremAngles around O add to 360° :

$$\angle AOC = 360^\circ - 108^\circ - 124^\circ = 128^\circ$$

5. Use trigonometrySince $OA = OC$,

$$\angle CAO = \frac{180^\circ - 128^\circ}{2} = 26^\circ$$

6. (c) Angle stands on the(c) Angle ABC stands on the major arc AC , so

$$\angle ABC = \frac{360^\circ - 128^\circ}{2} = 116^\circ$$

FINAL ANSWER $\angle AOD = 108^\circ$, $\angle CAO = 26^\circ$, and $\angle ABC = 116^\circ$.

PAST PAPER QUESTION**Q#: 17****Marks: 3**TOPIC: **Vectors**

Nov 2024 P2H - Nov 2024 | P2H

17 Here are two vectors.

$$\vec{FG} = \begin{pmatrix} -5 \\ 2 \end{pmatrix} \quad \vec{HG} = \begin{pmatrix} 4 \\ 14 \end{pmatrix}$$

Calculate the magnitude of the vector $\vec{HF}$

(Total for Question 17 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 17****Marks: 3****TOPIC: Vectors**

Nov 2024 P2H - Nov 2024 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use vectors**

$$\overrightarrow{GF} = -\overrightarrow{FG} = \begin{pmatrix} 5 \\ -2 \end{pmatrix}$$

$$\overrightarrow{HF} = \overrightarrow{HG} + \overrightarrow{GF} = \begin{pmatrix} 4 \\ 14 \end{pmatrix} + \begin{pmatrix} 5 \\ -2 \end{pmatrix} = \begin{pmatrix} 9 \\ 12 \end{pmatrix}$$

2. Magnitude

Magnitude:

$$|\overrightarrow{HF}| = \sqrt{9^2 + 12^2} = 15$$

FINAL ANSWER

15.

PAST PAPER QUESTION**Q#: 18****Marks: 4****TOPIC: Linear Graphs ($y = mx + c$)**

Nov 2024 P2H - Nov 2024 | P2H

- 18** The straight line **L** is perpendicular to the straight line with equation $2x + y = 9$ and passes through the point with coordinates (8, 11)

Find an equation for **L**

Give your answer in the form $y = mx + c$

(Total for Question 18 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 18****Marks: 4****TOPIC: Linear Graphs ($y = mx + c$)**

Nov 2024 P2H - Nov 2024 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. The given line is**

The given line is

$$2x + y = 9$$

$$y = -2x + 9$$

2. Find the gradientIts gradient is -2 , so a perpendicular line has gradient $\frac{1}{2}$.**3. Find the gradient**Line L passes through $(8, 11)$:

$$y - 11 = \frac{1}{2}(x - 8)$$

$$y = \frac{1}{2}x + 7$$

FINAL ANSWER

$$y = \frac{1}{2}x + 7.$$

PAST PAPER QUESTION**Q#: 19****Marks: 3****TOPIC: Transformations of Graphs**

Nov 2024 P2H - Nov 2024 | P2H

19 A curve has equation $y = f(x)$

There is only one minimum point on the curve.

The coordinates of this minimum point are (5, 4)

Write down the coordinates of the minimum point on the curve with equation

(i) $y = f(x + 5)$

(.....,)
(1)

(ii) $y = 3f(x)$

(.....,)
(1)

(iii) $y = f(x) - 7$

(.....,)
(1)

(Total for Question 19 is 3 marks)

PAST PAPER SOLUTION**Q#: 19****Marks: 3****TOPIC: Transformations of Graphs**

Nov 2024 P2H - Nov 2024 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. The minimum point of is**The minimum point of $y = f(x)$ is

$$(5, 4)$$

2. Read the graph

(i) For

$$y = f(x + 5)$$

3. the graph moves 5 units

the graph moves 5 units left, so the minimum point becomes

$$(0, 4)$$

4. Read the graph

(ii) For

$$y = 3f(x)$$

5. the y coordinate is multiplied

the y-coordinate is multiplied by 3, so the minimum point becomes

$$(5, 12)$$

6. Read the graph

(iii) For

$$y = f(x) - 7$$

7. the graph moves 7 units

the graph moves 7 units down, so the minimum point becomes

$$(5, -3)$$

FINAL ANSWER

(0, 4), (5, 12), and (5, -3).

PAST PAPER QUESTION**Q#: 20****Marks: 3****TOPIC: Solving Inequalities**

Nov 2024 P2H - Nov 2024 | P2H

- 20** Solve the inequality $10x^2 + 11x - 21 < 0$
Show clear algebraic working.

(Total for Question 20 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 20****Marks: 3****TOPIC: Solving Inequalities**

Nov 2024 P2H - Nov 2024 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Solve inequality**

$$10x^2 + 11x - 21 < 0$$

2. Factorise

Factorise:

$$10x^2 + 11x - 21 = (10x + 21)(x - 1)$$

3. The critical values are

The critical values are

$$x = -\frac{21}{10} \quad \text{and} \quad x = 1$$

4. The quadratic is less than

The quadratic is less than 0 between the roots.

FINAL ANSWER

$$-\frac{21}{10} < x < 1$$

PAST PAPER QUESTION**Q#: 20****Marks: 3****TOPIC: Solving Inequalities**

Nov 2024 P2H - Nov 2024 | P2H

- 20** Solve the inequality $10x^2 + 11x - 21 < 0$
Show clear algebraic working.

(Total for Question 20 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 20****Marks: 3****TOPIC: Solving Inequalities**

Nov 2024 P2H - Nov 2024 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Solve inequality**

$$10x^2 + 11x - 21 < 0$$

2. Factorise

Factorise:

$$10x^2 + 11x - 21 = (10x + 21)(x - 1)$$

3. The critical values are

The critical values are

$$x = -\frac{21}{10} \quad \text{and} \quad x = 1$$

4. The quadratic is less than

The quadratic is less than 0 between the roots.

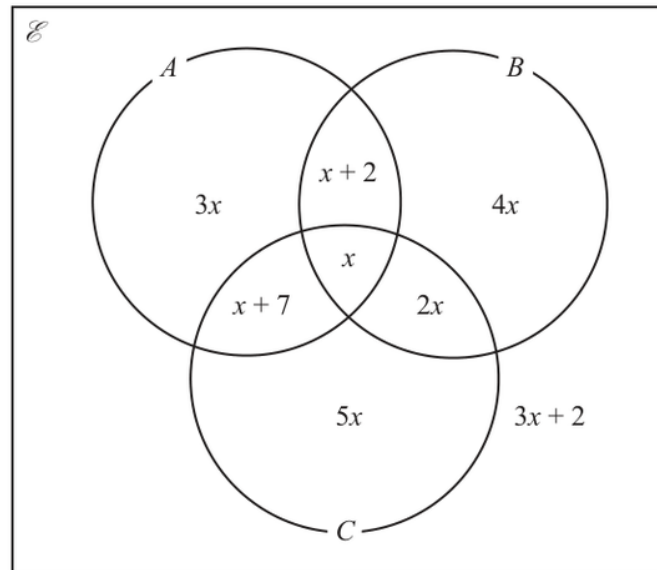
FINAL ANSWER

$$-\frac{21}{10} < x < 1$$

PAST PAPER QUESTION**Q#: 21****Marks: 4****TOPIC: Set Notation & Venn Diagrams**

Nov 2024 P2H - Nov 2024 | P2H

- 21** The Venn diagram shows information about the numbers of items in set A , set B and set C , where x is an integer.



Given that $n(A \cup B)' = 26$

find $n(A' \cap C)$

$$n(A' \cap C) = \dots\dots\dots$$

(Total for Question 21 is 4 marks)

PAST PAPER SOLUTION

Q#: 21

Marks: 4

TOPIC: **Set Notation & Venn Diagrams**

Nov 2024 P2H - Nov 2024 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. The complement of contains the

The complement of $A \cup B$ contains the regions outside both A and B . These are $5x$ and $3x + 2$.

2. Calculate value

So $5x + (3x + 2) = 26$, hence $8x + 2 = 26$ and $x = 3$.

3. Calculate statistic

Now $A' \cap C$ means inside C , but not inside A . These are the regions $5x$ and $2x$.

4. Calculate value

So $n(A' \cap C) = 5x + 2x = 7x = 7(3) = 21$.

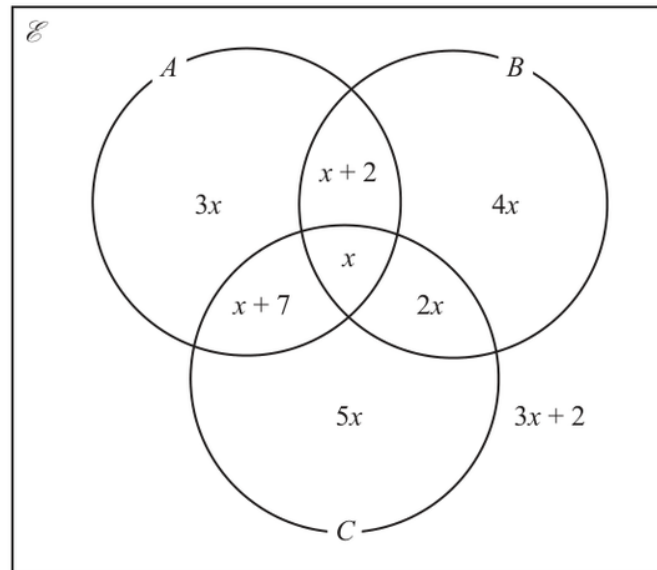
FINAL ANSWER

21

PAST PAPER QUESTION**Q#: 21****Marks: 4****TOPIC: Set Notation & Venn Diagrams**

Nov 2024 P2H - Nov 2024 | P2H

- 21** The Venn diagram shows information about the numbers of items in set A , set B and set C , where x is an integer.



Given that $n(A \cup B)' = 26$

find $n(A' \cap C)$

$$n(A' \cap C) = \dots\dots\dots$$

(Total for Question 21 is 4 marks)

PAST PAPER SOLUTION

Q#: 21

Marks: 4

TOPIC: **Set Notation & Venn Diagrams**

Nov 2024 P2H - Nov 2024 | P2H

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FINAL ANSWER

21

PAST PAPER QUESTION**Q#: 22****Marks: 5****TOPIC: Simultaneous Equations**

Nov 2024 P2H - Nov 2024 | P2H

22 Solve the simultaneous equations

$$x^2 + y^2 + y = 3$$

$$x + 2 = y$$

Show clear algebraic working.

(Total for Question 22 is 5 marks)

EA

PAST PAPER SOLUTION**Q#: 22****Marks: 5****TOPIC: Simultaneous Equations**

Nov 2024 P2H - Nov 2024 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Calculate value**

$$x^2 + y^2 + y = 3$$

$$y = x + 2$$

2. Substitute

Substitute:

$$x^2 + (x + 2)^2 + (x + 2) = 3$$

$$x^2 + x^2 + 4x + 4 + x + 2 = 3$$

$$2x^2 + 5x + 3 = 0$$

$$(2x + 3)(x + 1) = 0$$

3. Evaluate fraction

So

$$x = -\frac{3}{2} \quad \text{or} \quad x = -1$$

4. FindFind y :

$$x = -\frac{3}{2} \Rightarrow y = \frac{1}{2}$$

$$x = -1 \Rightarrow y = 1$$

FINAL ANSWER

$$(x, y) = \left(-\frac{3}{2}, \frac{1}{2}\right) \text{ or } (-1, 1)$$

PAST PAPER QUESTION**Q#: 22****Marks: 5****TOPIC: Simultaneous Equations**

Nov 2024 P2H - Nov 2024 | P2H

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$$x^2 + y^2 + y = 3$$

$$x + 2 = y$$

Show clear algebraic working.

(Total for Question 22 is 5 marks)

EA

PAST PAPER SOLUTION**Q#: 22****Marks: 5****TOPIC: Simultaneous Equations**

Nov 2024 P2H - Nov 2024 | P2H

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FINAL ANSWER

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PAST PAPER QUESTION**Q#: 23****Marks: 4****TOPIC: Algebraic Proof**

Nov 2024 P2H - Nov 2024 | P2H

23 Show that $\frac{16x^2 - 36}{x - 7} \div \frac{2x^2 + 7x + 6}{x^2 - 5x - 14} - (7 + 8x) = n$

where n is an integer to be found.
Show clear algebraic working.

(Total for Question 23 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 23****Marks: 4****TOPIC: Algebraic Proof**

Nov 2024 P2H - Nov 2024 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Evaluate fraction**

$$\frac{16x^2 - 36}{x - 7} \div \frac{2x^2 + 7x + 6}{x^2 - 5x - 14} - (7 + 8x)$$

2. Factorise

Factorise:

$$16x^2 - 36 = 4(2x - 3)(2x + 3)$$

$$2x^2 + 7x + 6 = (2x + 3)(x + 2)$$

$$x^2 - 5x - 14 = (x - 7)(x + 2)$$

3. Evaluate fraction

So

$$\begin{aligned} & \frac{4(2x - 3)(2x + 3)}{x - 7} \div \frac{(2x + 3)(x + 2)}{(x - 7)(x + 2)} - (7 + 8x) \\ &= \frac{4(2x - 3)(2x + 3)}{x - 7} \times \frac{(x - 7)(x + 2)}{(2x + 3)(x + 2)} - (7 + 8x) \\ &= 4(2x - 3) - (7 + 8x) \\ &= 8x - 12 - 7 - 8x \\ &= -19 \end{aligned}$$

4. Calculate valueSo $n = -19$, which is an integer.**FINAL ANSWER**

$$n = -19$$

PAST PAPER QUESTION**Q#: 23****Marks: 4****TOPIC: Algebraic Proof**

Nov 2024 P2H - Nov 2024 | P2H

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(Total for Question 23 is 4 marks)

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Nov 2024 P2H - Nov 2024 | P2H

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$$n = -19$$

PAST PAPER QUESTION**Q#: 24****Marks: 6****TOPIC: Bearings, Scale Drawing & Constructions**

Nov 2024 P2H - Nov 2024 | P2H

24 A , B and C are three points on horizontal ground.

$AB = 8.4$ metres $BC = 9.2$ metres

B is on a bearing of 067° from A

C is on a bearing of 129° from B

Calculate the bearing of A from C

Give your answer correct to the nearest degree.

0

(Total for Question 24 is 6 marks)

Elite

PAST PAPER SOLUTION**Q#: 24****Marks: 6****TOPIC: Bearings, Scale Drawing & Constructions**

Nov 2024 P2H - Nov 2024 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Resolve each journey into east**

Resolve each journey into east and north components.

2. Use trigonometryFrom A to B :

$$\text{east} = 8.4 \sin 67^\circ = 7.7322 \dots$$

$$\text{north} = 8.4 \cos 67^\circ = 3.2821 \dots$$

3. Use trigonometryFrom B to C :

$$\text{east} = 9.2 \sin 129^\circ = 7.1497 \dots$$

$$\text{north} = 9.2 \cos 129^\circ = -5.7897 \dots$$

4. Calculate valueSo from A to C :

$$\text{east} = 14.8820 \dots$$

$$\text{north} = -2.5076 \dots$$

5. Calculate valueTherefore from C to A :

$$\text{east} = -14.8820 \dots, \quad \text{north} = 2.5076 \dots$$

6. Use trigonometry

This direction is west of north.

$$\theta = \tan^{-1} \left(\frac{14.8820 \dots}{2.5076 \dots} \right) = 80.4355 \dots^\circ$$

7. Use bearingsBearing of A from C :

$$360^\circ - 80.4355 \dots^\circ = 279.5644 \dots^\circ$$

FINAL ANSWER

280°

EA

PAST PAPER QUESTION**Q#: 24****Marks: 6****TOPIC: Bearings, Scale Drawing & Constructions**

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PAST PAPER QUESTION**Q#: 25****Marks: 5****TOPIC: Circles, Arcs & Sectors**

Nov 2024 P2H - Nov 2024 | P2H

25 The diagram shows an equilateral triangle ABC and a circle with centre O

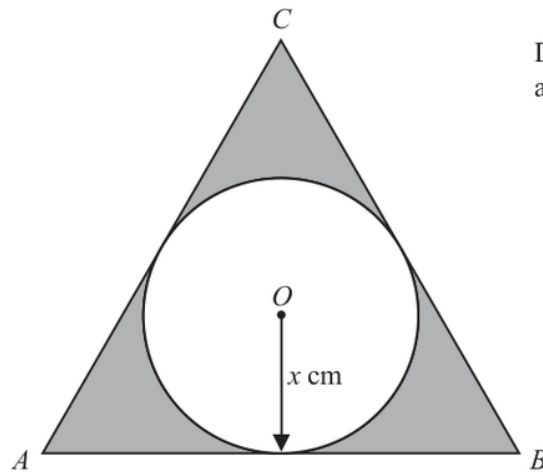


Diagram **NOT**
accurately drawn

AB , BC and CA are tangents to the circle.

The radius of the circle is x cm

The total area, in cm^2 , of the regions shown shaded in the diagram is nx^2

Find the value of n

Give your answer correct to 3 significant figures.

$n = \dots\dots\dots$

(Total for Question 25 is 5 marks)

PAST PAPER SOLUTION**Q#: 25****Marks: 5****TOPIC: Circles, Arcs & Sectors**

Nov 2024 P2H - Nov 2024 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use trigonometry**

The circle has radius x , so the inradius of the equilateral triangle is x .

2. Use trigonometry

For an equilateral triangle with side length a ,

$$\text{inradius} = \frac{a\sqrt{3}}{6}$$

3. Simplify surd

So

$$x = \frac{a\sqrt{3}}{6}$$

$$a = \frac{6x}{\sqrt{3}} = 2\sqrt{3}x$$

4. Area of the triangle

Area of the triangle:

$$\begin{aligned} \frac{\sqrt{3}}{4}a^2 &= \frac{\sqrt{3}}{4}(2\sqrt{3}x)^2 \\ &= \frac{\sqrt{3}}{4}(12x^2) = 3\sqrt{3}x^2 \end{aligned}$$

5. Area of the circle

Area of the circle:

$$\pi x^2$$

6. The shaded area is

The shaded area is

$$\begin{aligned} 3\sqrt{3}x^2 - \pi x^2 \\ = (3\sqrt{3} - \pi)x^2 \end{aligned}$$

7. Simplify surd

Therefore

$$n = 3\sqrt{3} - \pi$$

FINAL ANSWER

$$3\sqrt{3} - \pi$$

EA

PAST PAPER QUESTION

Q#: 25

Marks: 5

TOPIC: **Circles, Arcs & Sectors**

Nov 2024 P2H - Nov 2024 | P2H

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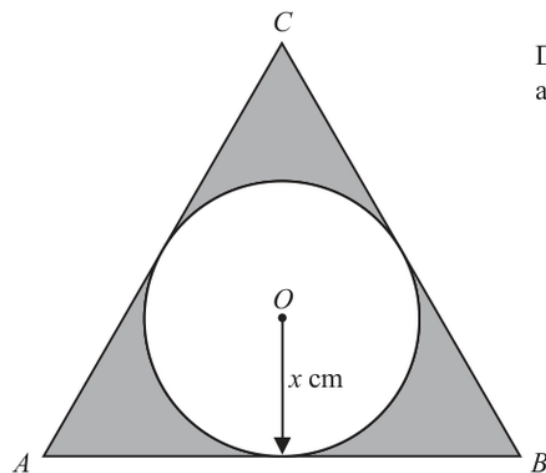


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7. Simplify surd

Therefore

$$n = 3\sqrt{3} - \pi$$

FINAL ANSWER

$$3\sqrt{3} - \pi$$

EA